BAYESIAN MODELING OF COMPLEXITY

ABM II: Implementation & Analysis

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Agenda

1 Recap & Today

2 Pattern-Oriented Modeling (POM)

3 Model Analysis

4 Course Project

5 Wrap-Up & Next

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Last Session: What We Built

Completed components:

- ✓ Understood ODD protocol structure
- ✓ Learned Design Concepts (emergence, sensing, etc.)
- ✓ Implemented basic Schelling structure:
  - `Grid` class with neighbor detection
  - `Agent` class with happiness checking
  - `SchellingModel` initialization

What's still missing:

- Movement logic (agents relocating)
- Simulation loop (time steps)
- Stopping conditions
- Visualization & metrics
- Analysis & interpretation

Today's Learning Goals

By the end of this session, you will be able to:

1. **Apply** Pattern-Oriented Modeling (POM):
   - Identify characteristic patterns
   - Validate model against patterns
2. **Analyze** model behavior:
   - Parameter sweeps
   - Sensitivity analysis
   - Verification vs. validation
3. **Start** thinking about your ABM project

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What is Pattern-Oriented Modeling?

Definition

Pattern-Oriented Modeling (POM) is the scientific method applied to ABM development using multiple patterns observed in real systems to guide model structure and parameterization.

Core idea:

- Real systems exhibit characteristic patterns at multiple levels
- ABM should reproduce these patterns if mechanisms are correct
- Patterns constrain both **structure** (what to include) and **parameters** (how to set values)

The POM Workflow

1. **Identify characteristic patterns** from real system
   - Multiple patterns at different scales/levels, e.g., individual, population, spatial, temporal
2. **Build initial model** (minimal/simple version)
   - Start with essential mechanisms only
3. **Test pattern reproduction**
   - Does model generate observed patterns?
   - Define matching criteria (quantitative when possible)
4. **Diagnose mismatches**
   - Which patterns fail? Why?
   - What's missing from model structure?
5. **Revise model** to enable pattern emergence
   - Add mechanisms, adjust parameters
6. **Iterate** until satisfactory pattern matching achieved

Goal: Decode internal organization of real system through iterative refinement

Types of Patterns

Patterns can appear at multiple levels:

Individual-level:

- Behavioral rules (decisions)
- Movement patterns

Population-level:

- Aggregate statistics
- Size/density patterns

Spatial:

- Clustering
- Dispersion patterns

Temporal:

- Dynamics trajectories
- Cycles/oscillations

Interaction:

- Network structure & patterns
- Competition/cooperation

Response:

- Sensitivity to perturbations
- Experimental outcomes

Weak Patterns are Informative Too!

Even patterns that show "no effect" or "weak correlation" constrain models, don't ignore them!

POM Applied to Schelling Model

Characteristic patterns in real urban segregation:

1. **High segregation levels:** 70-90% same-group neighbors in segregated cities
2. **Spatial clustering:** Continuous neighborhoods, not random scattering
3. **Persistence:** Once established, segregation is stable over time
4. **Mild preferences:** Survey data shows moderate tolerance (not extreme racism)
5. **Tipping points:** Rapid segregation after threshold neighborhood change
6. **Scale dependence:** Segregation metrics vary by geo scale

Model validation:

- Does Schelling reproduce these patterns?
- Which parameter values match observed segregation?
- What patterns does it *fail* to reproduce? why? (missing mechanisms)

Pattern Matching: Quantitative Criteria

How to define "good match"?

1. **Visual inspection:**
   - Qualitative comparison of spatial patterns
   - Quick, intuitive, but subjective
2. **Statistical distance:**
   - Euclidean distance: $d = \sqrt{\sum(y_{\text{model}} - y_{\text{data}})^2}$ where y denotes the pattern
   - Mean absolute error (MAE)
3. **Pattern-specific metrics:**
   - Segregation indices
   - Clustering coefficients
   - Distribution comparisons
4. **Qualitative criteria:**
   - Model produces increasing segregation
   - Convergence time is reasonable

For Schelling: Do real-world visual segregation patterns appear? Does segregation index match the observed (0.7-0.9)?

POM for Theory Development

Question: What agent behavior best explains observed patterns?

Scientific method applied to ABM (to test hypotheses):

1. **Devise alternative hypotheses** → Multiple candidate traits/rules for agents
2. **Design experiment to distinguish hypotheses** → Implement all candidate traits in model
3. **Carry out experiment** → Test which trait reproduces characteristic patterns
4. **Exclude non-performing hypotheses**
5. **Repeat** → With refined hypotheses

Important: We don't want to reproduce *all* agent behaviors, only those **relevant to the question**!

Example

Can the segregation be due to agents moving randomly and purposefully? Test both random and purposeful, e.g., moving to the closest empty cell and compare against patterns.

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How to Analyze ABMs?

Running the model once is not enough!

We need to understand:

1. **Robustness:** Do results hold across parameter ranges?
2. **Sensitivity:** Which parameters matter most?
3. **Uncertainty:** How much is the effect of stochastic elements?
4. **Mechanisms:** What drives the observed patterns?
5. **Generalizability:** Do findings apply beyond specific settings?

Key analysis techniques:

- Parameter sweeps & Sensitivity analysis
- Monte Carlo replication (stochastic models)
- Verification & validation

Verification vs. Validation

Verification:

Are we building the model *right*?

- Is the code bug-free?
- Implementation matches design?
- Are the algorithms correct?

Methods:

- Unit tests + code reviews
- Visualization checks
- Boundary condition tests

Validation:

Are we building the *right* model?"

- Does model represent reality?
- Do outputs match real data?
- Are mechanisms plausible?

Methods:

- Pattern matching (POM)
- Comparison to data
- Out-of-sample prediction

Key Distinction

Verification: Technical correctness | Validation: Scientific credibility

Both are necessary! A correctly coded wrong model is useless. A great model with bugs is also useless.

Verification Techniques for Schelling

How to verify our implementation:

1. **Visualization checks:**
   - Watch animation: do agents move sensibly?
   - Are grid updates correct after moves?
2. **Sanity checks:**
   - Number of agents constant over time?
   - Metrics in valid range (0-1)?
   - Grid occupancy = number of agents?
3. **Boundary conditions:**
   - Tolerance = 0%: All unhappy → maximum movement
   - Tolerance = 100%: All happy immediately
   - Density = 100%: No empty cells → no movement
4. **Known results:**
   - Segregation should increase (not decrease)
   - Eventually converges or reaches max steps

Exercise: Test tolerance = 0 and tolerance = 1. Do results make sense?

Parameter Sweeps

Goals:

- Understand how outcomes vary with parameters
- Identify thresholds/tipping points
- Find parameter ranges that match observations

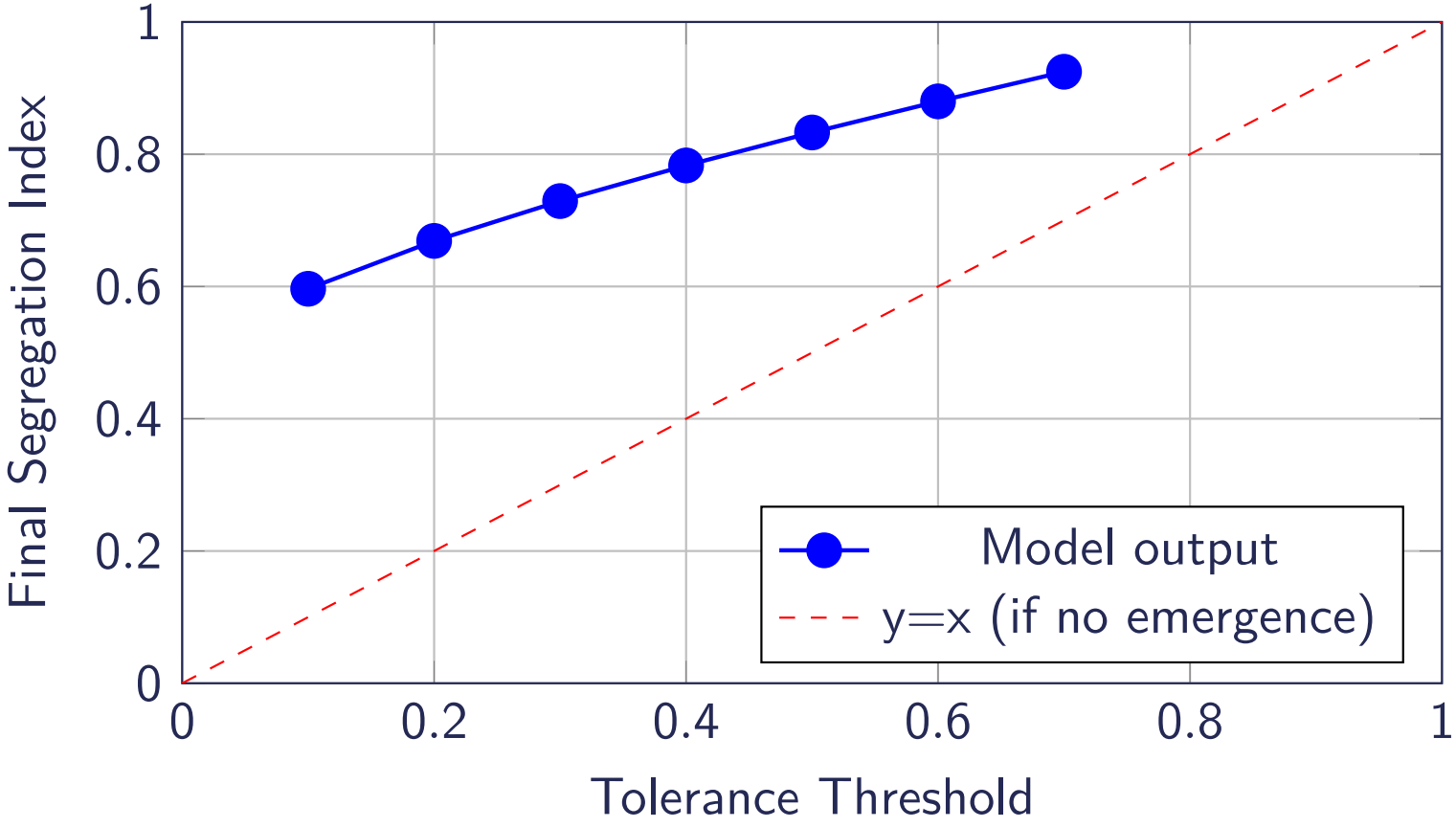
Approach for Schelling:

1. Choose parameter to vary (e.g., tolerance: 0.1 to 0.7)
2. Run model for each value (multiple replicates)
3. Record outcome metrics (final segregation, convergence time)
4. Plot relationships

Expected Pattern

Hypothesis: Higher tolerance $\rightarrow$ lower segregation
Test: Sweep tolerance from 0.1 to 0.7 in steps of 0.1
Result: Plot segregation index vs. tolerance

Example: Tolerance Sweep Results



Interpretation: Even mild tolerance (30%) produces strong segregation (75%). Could it be **emergence**!

Sensitivity Analysis

Goal: Identify which parameters have largest impact on outcomes

Why important?

- Focus calibration efforts on sensitive parameters
- Understand model behavior

Types of sensitivity analysis:

1. **Local sensitivity:**
   - Vary one parameter at a time
   - Keep others at baseline
   - Measure output change
2. **Global sensitivity:**
   - Vary all parameters simultaneously (sampling)
   - Captures interactions between parameters

Monte Carlo Replication

Problem: ABMs with random elements give different results each run

Solution: Multiple replications with different random seeds, **How?**

1. Fix parameter values
2. Run model N times (e.g., $N = 50$) with different random seeds
3. Record outcome for each run
4. Compute statistics: mean, std dev, confidence intervals

For Schelling:

- Stochastic elements: Initial placement, move order, destination
- Expect some run-to-run variation in final segregation
- Report: "Mean segregation $= 0.76 \pm 0.03$ (SD) across 50 runs"

Rule of Thumb

For stochastic models, always report results from multiple runs (not just one)!

Parameter Analysis

Task: Analyze how TWO parameters affect segregation

Instructions:

1. **Choose 2 parameters to vary:**
   - Tolerance (0.2, 0.3, 0.4, 0.5)
   - Density (0.6, 0.7, 0.8, 0.9)
2. **Run experiments:**
   - For each parameter combination
   - Run N replicates per combination
   - Record final segregation index
3. **Visualize results:**
   - Create 2D heatmap or line plot
   - Show mean $\pm$ standard deviation
4. **Interpret:** Which parameter matters more?

Hint: Nested loops over parameter values, collect results in array

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ABM Course Project

Your task: Design, implement, and analyze your own ABM

Requirements:

1. **Topic:** Choose a complex system (social, economic, etc.)
2. **Documentation:** Write ODD protocol (all 3 sections)
3. **Implementation:** Working Python code (OOP structure)
4. **Analysis:**
   - Parameter sweep on the parameters
   - Pattern-oriented validation
   - Visualization (spatial + time series)
5. **Bayesian component:** Apply technique from Sessions 4-11:
   - Calibrate model using ABC, OR
   - Implement Bayesian agents, OR
   - Analyze outputs with Bayesian regression
6. **Presentation:** 10 min (Session 12)
7. **Code submission:** Github repo

Project Ideas & Scope

Good project characteristics:

- Clear question (testable hypothesis)
- 2-4 agent types or behaviors
- Spatial or network structure (optional but recommended)
- Observable emergent pattern
- Manageable scope (you have ∼7 weeks)

Example topics:

- **Social:** Opinion dynamics, cooperation, social networks, voting
- **Economic:** Market dynamics, Investor risk-profit
- **Ecological:** Predator-prey, resource competition, dispersal
- **Epidemiological:** Disease spread, vaccination strategies
- **Urban:** Traffic flow, land use, gentrification
- **Political:** Coalition formation, conflict, policy diffusion

Start Simple!

Build minimal version first, then add complexity incrementally. Better to have working simple model than broken complex one.

Project Timeline & Milestones

Session	Milestone
4	Topic selection, group formation
7	Early design submission, ODD documentation, preliminary analysis
8-11	Add Bayesian component, complete analysis, parameter sweeps, finalize code, create visualizations
12	**Presentations** (10 min + 5 min Q&A)

Deliverables:

- ODD document (PDF)
- GitHub with well-structured code and README.md with proper instructions to run the model
- Presentation slides

Grading: 25% of final grade (see rubric on Moodle)

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What We Accomplished Today

- ✓ **Completed Schelling implementation:**
  - Movement logic
  - Simulation loop
  - Metrics tracking
  - Stopping conditions
- ✓ **Learned visualization techniques:**
  - Spatial grid plots
  - Time series metrics
- ✓ **Understood Pattern-Oriented Modeling:**
  - Using patterns to guide development
  - Validation vs. verification
- ✓ **Analyzed model behavior:**
  - Parameter sweeps
  - Sensitivity concepts
  - Monte Carlo replication
- ✓ **Project assigned:** Start thinking about your system!

Next Session: Introduction to Bayesian Statistics

Session 4: Probability Foundations & Bayes' Law

Topics:

- Random variables, PDF, CDF
- Joint and conditional probability
- Bayes' Law: $P(\theta|D) \propto P(D|\theta)P(\theta)$
- Classic examples (medical testing, false positives)
- Hands-on calculations

Why now?

- We have working ABMs with unknown parameters
- Need statistical tools to learn from data
- Bayesian methods are ideal for ABM calibration (Session 8!)

Transition:

From building models → to learning from data

Homework & Preparation

Before next session:

1. **Complete Schelling analysis:**
   - Run full parameter sweep (download notebook from Moodle)
   - Create all visualizations
   - Write 1-paragraph interpretation of results
2. **Start project planning:**
   - Choose topic (or 2-3 candidates)
   - Draft Purpose section of ODD
   - Identify 2-3 characteristic patterns to reproduce
3. **Optional reading:**
   - Schelling (1971) original paper
   - Railsback & Grimm Chapter 9 (POM)
4. **Review basic probability:**
   - Random variables, distributions
   - Conditional probability
   - We'll build on this next session

Questions?

Next session: Entering the Bayesian world!